Ziyi Sun

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EDUCATION

Ocean University of China, Qingdao China **2023-Present**

B.Eng. Candidate in Mechanical Design, Manufacturing and Automation Sep. 2023 to Jun. 2027 Expected

- **GPA:** 3.7/4.0 (rank:4/56)
- **Selected Major Course Grade:** Advanced Mathematics (97) Linear Algebra (93.5) Probability Theory and Mathematical Statistics (93) Complex Variables and Integral Transforms (94.5) Numerical Methods and Applications (96) Theory of Machines and Mechanisms (89) Fundamentals of Mechanical Vibrations (95) Engineering Testing Technology (93) Mechanics of Materials (91) Engineering Materials and Forming Technology (89.9)

RESEARCH EXPERIENCE

[1] A Two-Stage Learning Framework for Autonomous Obstacle Avoidance of an Eel-Like Robot

IEEE RCAR 2026

__Ziyi Sun__[2], Luwen Li[2], Changhong He[2], Shuo Jiang[2], Zhuoyan Wang[2], Haoyuan Cheng[1,2,] ([1]State Key Laboratory of Intelligent Coal Mining and Strata Control, Beijing, China;[2]College of Engineering, Ocean University of China, Qingdao 266100, China;[*]Corresponding author)*

- Developed a vision-based robot learning framework for autonomous obstacle avoidance of an eel-like underwater robot, using imitation learning for policy initialization, PPO for fine-tuning, and visual domain randomization for sim-to-real transfer.
- Built a MuJoCo simulation environment with simplified underwater dynamics and an AHTA-CPG based action abstraction for strongly coupled eel-like locomotion, and validated the learned policy in 60 real-world water-tank trials, achieving 70% to 90% success rates across multiple obstacle configurations.

[2] Design of Biomimetic Robotic Eel with Omnidirectional Quadruple-Wire-Driven Body and Controlled Central Pattern Generator

ICCCR 2025, DOI: 10.1109/ICCCR65461.2025.11072624

Dingyang Yu, Yuanrong Zhong, Qi Chen[*], Siming Ke, Qianbei Luo, __Ziyi Sun__, Xunqi Zhao ([*]Corresponding author)

- Presented the mechanical design and control framework of a biomimetic robotic eel, integrating an omnidirectional quadruple wire-driven body with a controlled central pattern generator for flexible underwater locomotion.

[3] High-Stability PTO System for Duck-Type Wave Energy Converters

Invention Patent

Zhi Han[*], __Ziyi Sun__, Haoran Chen, Hailong Zhao, Wenchi Wei, Xueyao Wang, Hongda Shi[*] ([*]Corresponding author)

- Designed a high-stability power take-off system for duck-type wave energy converters, focusing on improving energy conversion stability under variable wave conditions.
- Contributed to the structural design and stability optimization of the PTO system, aiming to enhance the reliability and adaptability of duck-type wave energy conversion devices.

[4] Underwater Polarized Light Field Prediction Software for Biomimetic Navigation V1.0 Software Copyright

China National Copyright Administration, Registration No. 2024SR0140235

- Developed software for underwater polarized light field prediction, supporting biomimetic navigation, underwater perception analysis, and navigation cue modeling.

[5] Variable-Length Biomimetic Robotic Fish with Multimodal Perception for Deep-Sea Exploration

Student Research Development Program (SRDP), Ocean University of China

__Ziyi Sun__, Zhuoyan Wang, Xunqi Zhao, Yalin Liu, Shuo Jiang, Yingchun Xie[*], Qi Chen[*] ([*]Advisor)

- Led the design of a variable-length biomimetic robotic fish for adaptive operation in unstructured underwater environments, integrating modular antagonistic continuum units for morphology adaptation.
- Developed multimodal data fusion and motion performance evaluation methods using Kalman filtering and AHP, with responsibilities covering decision-system design, mechanical design, and motion simulation.

[6] Cross-Domain Biomimetic Eel Robot Based on a Wire-Driven Mechanism

Provincial Innovation and Entrepreneurship Training Project

***Ziyi Sun**, Dingyang Yu, Yuanrong Zhong,Siming Ke, Qianbei Luo, Xunqi Zhao, Qi Chen* (*Advisor)*

- Designed a cross-domain biomimetic eel robot for dynamic submarine cable inspection in offshore floating wind power systems, integrating wire-driven locomotion with buoyancy-gravity center regulation for efficient cruising and dual-mode surfacing/diving.
- Integrated visual recognition and Gaussian splatting for underwater cable reconstruction and damage detection, with responsibilities covering mechanical design and hydrodynamic analysis.

[7] High-Stability Duck-Type Wave Energy Converter Based on a Pendulum PTO System

National Innovation and Entrepreneurship Training Project

*Hailong Zhao, **Ziyi Sun**, Haoran Chen, Hailong Zhao, Wenchi Wei, Xueyao Wang, Zhi Han*, Hongda Shi* (*Advisor)*

- Optimized a Salter-D0018 duck-type wave energy converter to improve energy absorption efficiency and operational reliability.
- Used AQWA and genetic algorithms to optimize the hull profile and enhance wave energy capture.
- Developed a pendulum-based PTO design and adaptive wave-condition control strategy to improve power generation stability; responsible for control strategy design, mechanical design, and motion simulation.

[8] Modular Adaptive Deep-Sea Resource Collection and Locomotion Platform

Faculty-Led Industry Collaboration Project with China Merchants Group

*Luwen Li, **Ziyi Sun**, Zhuoyan Wang, Zicong Zhang, Shimin Yu*(*Advisor)*

- Designed a modular deep-sea mining vehicle with an active four-track locomotion system for complex seabed terrain.
- Used Fluent to analyze mining-induced plume generation and developed plume suppression structures with a compliant collection mechanism.

HONORS & AWARDS

- **Shandong Provincial Government Scholarship(2025)**
- **First-Class Comprehensive Scholarship, Ocean University of China(2024&2025)**
- **Outstanding Student, Ocean University of China(2025)**
- **Outstanding Student Cadre, Ocean University of China(2024)**
- **National First Prize, National 3D Digital Innovation Design Competition(2025)**
- **National Second Prize, Contemporary Undergraduate Mathematical Contest in Modeling(2025)**
- **Honorable Mention, Mathematical Contest in Modeling(2025)**
- **National Third Prize, 18th National College Student Energy Conservation and Emission Reduction Social Practice and Science & Technology Competition(2026)**

SKILLS

- **Programming:** Python, C
- **Robot Learning:** Reinforcement Learning, Imitation Learning, Sim-to-Real Transfer
- **Simulation & Modeling:** MuJoCo, MATLAB
- **Computer Vision:** Visual Recognition, Gaussian Splatting, Visual Domain Randomization
- **Robotics & Control:**Motion Simulation, Embedded Control, Mechatronic Debugging
- **Mechanical Design:** SolidWorks, 3D Modeling, Structural Design
- **Languages: Mandarin Chinese**, native; **English**, CET-6: 482, CET-4: 591; **German**, elementary coursework(Grade:87); **Italian**, elementary with early exposure

LEADERSHIP & SERVICE

- **Class Monitor,** Ocean University of China
- **Team Leader,** Engineering College Debate Team
- **Director,** Student Association for the Study of Socialism with Chinese Characteristics, Ocean University of China
- **Team Leader,** win Outstanding Individual in Summer Social Practice, Ocean University of China
- **Team Leader,** Led the team to win Award Science and Technology Innovation Pioneer Team, College of Engineering